Surgical Repair of Congenital Diaphragmatic Hernia After Extracorporeal Membrane Oxygenation Cannulation

Early Repair Improves Survival

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Objective: To determine the optimal timing of congenital diaphragmatic hernia (CDH) repair after extracorporeal membrane oxygenation (ECMO) cannulation

Summary Background Data: The timing of CDH repair after ECMO cannulation remains a controversial topic due to studies with low power or strong selection bias.

Methods: This is a 2-aim retrospective cohort study based on the CDH Study Group registry for the period of 2007–2017. Aim 1—Compare On versus After ECMO repair. Aim 2—Compare Early versus Late repair on ECMO. In order to minimize selection bias and account for non-repairs, subjects in each aim were stratified into study groups based on their treatment center's characteristics. In each aim, the study groups were matched based on propensity score (PS). The main outcomes included mortality rate and incidence of non-repair.

Results: In aim 1, 136 patients remained in each group after PS matching. Compared to the After ECMO group, patients in the On ECMO group demonstrated a lower mortality rate, hazard ratio (HR) 0.54 (0.38, 0.77) ($P < 0.001$), and lower incidence of non-repair, 5.9% versus 33.8% ($P < 0.001$). In aim 2, 77 patients remained in each group after PS matching. Compared to the Late group, Early repair of CDH on ECMO was associated with a lower mortality rate, HR 0.51 (0.33, 0.77) ($P = 0.002$), and lower incidence of non-repair, 9.1% versus 44.2% ($P < 0.001$).

Conclusions: The approach of early repair after ECMO cannulation is associated with improved survival compared to delayed surgical correction.

Keywords: congenital diaphragmatic hernia, extracorporeal membrane oxygenation, propensity score matching

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Congenital diaphragmatic hernia (CDH) is characterized by a defect in the diaphragm with herniation of abdominal organs into the thoracic cavity, pulmonary hypoplasia, and pulmonary hypertension.[1] Despite advancements in neonatal intensive care, CDH continues to carry a high mortality rate.[2,3] A widely used therapy in the management of CDH is extracorporeal membrane oxygenation (ECMO), which is employed in an average of 30% of patients and provides cardiopulmonary support for the most severe cases.[4,5] For CDH infants who require ECMO support, a troubling trend of decreased survival rates has been observed in recent years[6] and the optimal time to undertake surgical repair remains controversial.

Published studies on this topic have varying conclusions. The first report of repair on ECMO has a high rate of bleeding complications and an overall survival of 43%.[7] A single institutional report demonstrated a survival advantage with early repair on ECMO, but the small cohort of patients precluded performing a regression analysis.[8] Conversely, another single institutional report found an inverse relationship where early repair resulted in higher mortality and bleeding complications.[9] In one of the largest studies to date, Bryner et al[10] identified a strong survival advantage with repair after ECMO decannulation compared to repair on ECMO. However, the exposure classification in this study (repair on ECMO vs after ECMO) led to the exclusion of a large number of patients who were cannulated to ECMO but did not undergo repair, and therefore died. Therefore, selection bias was unaccounted for in this analysis.

In this study, we assessed the relationship between surgical timing after ECMO cannulation and survival, using both propensity score matching and a rigorous approach to study group classification, which allowed for the inclusion of patients who did not undergo surgical repair.

METHODS

We performed a retrospective cohort study using data from the CDH Study Group (CDHSG) registry. Founded in 1995, this is a voluntary international registry that collects data on liveborn infants with CDH from participating centers (Appendix). Patient data was collected from the time of birth to death or discharge from the index hospitalization. The study included 2 aims: comparing the survival of those repaired on ECMO versus after ECMO and comparing the survival of those who underwent early versus late repair on ECMO.

The study was approved by the Institutional Review Boards at Boston Children's Hospital (#M05-10-244) and the University of Texas – Houston (#HSC-MS-03-223). A 2-sided P value of 0.05 was used as the cut-off for statistical significance. All analyses were performed with SAS 9.4 (SAS Institute, Cary, NC).

Aim 1—On Versus After ECMO Repair

Primary Hypothesis

We hypothesized that surgical repair of CDH on ECMO would result in better survival compared to after-ECMO repair.

Inclusion and Exclusion Criteria

All patients born between January 1, 2007 and December 31, 2017 and cannulated for ECMO were included. Patients undergoing repair before ECMO or during a second run of ECMO were excluded.

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Study Group Classification

In order to account for patients who were cannulated to ECMO but failed to undergo surgical repair (non-repairs), we elected to not use the actual timing of surgery as exposure. Instead, subjects were stratified into quartiles based on their center's frequency of on-ECMO repair, defined as the number of on-ECMO repairs divided by the total ECMO cases over the study period. The highest quartile represented patients treated at centers that frequently perform repair on ECMO. This group was labeled On ECMO. In contrast, the After ECMO group included patients in the lowest quartile, those treated at centers that rarely repair on ECMO. Subjects in the middle 2 quartiles belonged to the Mixed group.

Outcomes and Analyses

Before Propensity Score (PS) Matching. Demographic and baseline characteristics were described for the 3 study groups. The primary outcome was mortality by discharge. Adjusted comparison of mortality rate was achieved with a multivariable Cox proportional hazards regression model, with After ECMO as the referent group. Confounding variables for regression models were chosen *a priori* and included cardiac and chromosomal anomalies, treatment center volume, inborn status, ECMO mode, prenatal diagnosis, pulmonary hypertension status, birth weight, EGA, and Apgar score at 5 minutes.[6,11–15] Some of these factors have been shown to have predictive properties for mortality, such as birth weight, Apgar score at 5 minutes, chromosomal or major cardiac abnormality, and suprasystemic pulmonary hypertension.[16]

After PS Matching. The On ECMO and After ECMO groups were matched based on PS. In order to perform PS matching, a logistic regression model using all demographic and baseline covariates (Tables 1 and 2) was first fit to generate a PS for each patient. This PS represented the probability of being assigned to one of the study groups given the patient's demographic and baseline characteristics. Subjects from one study group were sequentially matched with those in the other study group on a 1:1 ratio and without replacement. The matching method was nearest-neighbor. Matching was only allowed when the pair's caliper was within 25% of the pooled standard deviation of the logit of PS.

Analysis of primary outcomes was repeated on the matched cohorts. Survival analysis was achieved with Kaplan–Meier curves and a Cox regression model. Patients were censored at the time of hospital discharge or transfer, and time to event was measured from ECMO cannulation to death. Median survival time was determined from Kaplan–Meier curves and comparison of survival probability was performed with a Wilcoxon-weighted log-rank test. Additionally, in order to account for the possible correlation in outcomes among patients treated at the same center, a mixed effects logistic regression model was also used to compare the odds of death between the 2 matched groups. In this model, patients were clustered by treatment centers, exposure group was assumed to have a fixed effect, and the intercept was allowed to have a random effect.

The secondary outcomes included incidence of non-repair, length of ECMO, length of hospitalization, and incidence of oxygen dependence and oral feeding at discharge. Oxygen dependence referred to the need for mechanical ventilation, continuous positive airway pressure, or nasal cannula support. Oral feeding was defined as tolerance of more than 50% of nutrition by mouth. Comparison of the incidence of non-repair between the 2 matched study groups was achieved with a chi-square test and a Wilcoxon rank-sum test was used for ECMO duration. Among patients who survived to discharge, comparison of hospitalization duration was achieved with a Wilcoxon rank-sum test and a chi-square test was used for comparing the incidence of oxygen dependence and oral feeding at discharge.

Sensitivity Analysis. In order to reconcile the findings in this study with previously published results on the timing of repair and mortality, a sensitivity analysis was performed with exclusion of non-repairs. Comparison of mortality rate among the 3 study groups was repeated. Due to the reduced sample size, the number of covariates in the Cox proportional hazards regression model was reduced a priori to include cardiac anomalies, treatment center volume, inborn status, prenatal diagnosis, birth weight, and Apgar score at 5 minutes.

Aim 2—Early Versus Late Repair on ECMO

Secondary Hypothesis

We hypothesized that early surgical repair of CDH on ECMO would result in better survival compared to late repair on ECMO.

Inclusion and Exclusion Criteria

All patients born between January 1, 2007 and December 31, 2017 and cannulated to ECMO were included. Patients undergoing CDH repair before ECMO, after ECMO, or during a second run of ECMO were excluded. In addition, non-repairs who were treated at centers that never repair on ECMO were also excluded as these patients were deemed inappropriate for the second aim's analysis.

Study Group Classification

Similar to aim 1, the need to account for non-repairs precluded the use of individual surgical timing as the classification scheme. Instead, study subjects were classified based on their center's average duration to repair over the study period, defined as the time of ECMO cannulation to the time of surgical repair. Patients were again stratified into quartiles based on their center's average duration to repair. The highest quartile represented patients treated at centers that mainly repair late on ECMO and this group was labeled Late. In contrast, those in the lowest quartile belonged to the Early group. The remaining patients in the middle 2 quartiles were in the Mixed group.

Outcomes and Analyses

Before PS Matching. The primary outcome was mortality by discharge. Adjusted comparison of mortality rate was performed with a multivariable logistic and Cox regression model as previously described.

After PS Matching. Patients in the Early and Late groups were PS matched as previously mentioned. Comparison of survival was achieved with Kaplan–Meier analysis, Cox regression, and mixed effects logistic regression as previously described. Comparisons of secondary outcomes were performed with chi-square and Wilcoxon rank-sum tests as previously described.

Sensitivity Analysis. Similar to aim 1, a sensitivity analysis was performed with exclusion of non-repairs. Comparisons of mortality rate among the 3 study groups were repeated with a Cox regression model as previously mentioned.

RESULTS

Aim 1—On Versus After ECMO Repair

From 2007 to 2017, 1581 patients with CDH were cannulated to ECMO. After excluding 147 patients who were repaired before ECMO cannulation and 11 patients who were repaired on a second run of ECMO, there were 1423 patients in the primary analysis. The

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TABLE 1. Aim 1—On Versus After ECMO Repair

	Before PS Matching				After PS Matching		
	After ECMO (N = 360)	Mixed (N = 696)	On ECMO (N = 367)	P	After ECMO (N = 136)	On ECMO (N = 136)	P
Male	222 (61.7%)	394 (56.7%)	216 (59.0%)	0.29	81 (59.6%)	86 (63.2%)	0.53
Race/ethnicity				< 0.001			0.49
White	128 (38.4%)	433 (68.7%)	238 (73.0%)		65 (47.8%)	68 (50.0%)	
Black	56 (16.8%)	101 (16.0%)	27 (8.3%)		14 (10.3%)	19 (14.0%)	
Other	149 (44.7%)	96 (15.2%)	61 (18.7%)		57 (41.9%)	49 (36.0%)	
Inborn	160 (44.7%)	364 (52.3%)	261 (71.1%)	< 0.001	79 (58.1%)	80 (58.8%)	0.90
High-volume center	193 (53.6%)	521 (74.9%)	271 (73.8%)	< 0.001	88 (64.7%)	80 (58.8%)	0.32
Birth weight (kg)	3.12 ± 0.51	3.02 ± 0.51	3.04 ± 0.53	0.006	3.10 ± 0.47	3.08 ± 0.55	0.75
EGA (wk)	38.1 ± 1.6	37.8 ± 1.6	37.9 ± 1.9	0.08	38.2 ± 1.5	38.1 ± 1.8	0.74
Prenatal diagnosis	261 (72.5%)	550 (79.0%)	283 (77.1%)	0.06	99 (72.8%)	94 (69.1%)	0.59
APGAR							
1 min	3.9 ± 2.2	3.9 ± 2.2	3.9 ± 2.3	0.94	3.9 ± 2.2	3.9 ± 2.1	0.76
5 min	6.1 ± 2.0	5.9 ± 2.1	5.7 ± 2.2	0.03	6.0 ± 2.1	5.7 ± 2.2	0.19
Cardiac anomaly				0.18			0.77
None	291 (80.8%)	519 (74.6%)	288 (78.5%)		106 (77.9%)	101 (74.3%)	
Minor	45 (12.5%)	113 (16.2%)	54 (14.7%)		22 (16.2%)	26 (19.1%)	
Major	24 (6.7%)	64 (9.2%)	25 (6.8%)		8 (5.9%)	9 (6.6%)	
Chromosomal anomaly	10 (2.8%)	43 (6.2%)	23 (6.3%)	0.04	5 (3.7%)	8 (5.9%)	0.57
Left side hernia	274 (76.8%)	558 (80.8%)	283 (78.6%)	0.30	99 (72.8%)	103 (75.7%)	0.68
PHTN degree				< 0.001			0.86
None	17 (5.3%)	36 (5.9%)	23 (6.8%)		7 (5.2%)	8 (5.9%)	
Mild	23 (7.1%)	53 (8.7%)	17 (5.1%)		11 (8.1%0	8 (5.9%)	
Moderate	100 (31.0%)	265 (43.7%)	119 (35.4%)		46 (33.8%)	50 (36.8%)	
Severe	183 (56.7%)	253 (41.7%)	177 (52.7%)		72 (52.9%)	70 (51.5%)	
ECMO mode				< 0.001			0.86
VV	97 (28.4%)	89 (13.2%)	30 (8.6%)		21 (15.4%)	19 (14.0%)	
VA	245 (71.6%)	585 (86.8%)	321 (91.4%)		115 (84.6%)	117 (86.0%)	

Demographic and baseline characteristics of the study groups before and after propensity score (PS) matching. High volume is defined as $\geq$ 10 cases of CDH per year. Cardiac anomalies are classified as minor or major. Minor cardiac anomalies include atrial and ventricular septal defect; major anomalies include diagnoses such as hypoplastic left heart syndrome, coarctation of the aorta, double outlet right ventricle, and congenital cyanotic heart diseases. Pulmonary hypertension (PHTN) is categorized as none, mild (less than 2/3 systemic pressure), moderate (between 2/3 and systemic pressure), and severe (higher than systemic pressure). Categorical variables are expressed as number and percentage. Continuous variables are expressed as mean $\pm$ standard deviation (SD). Comparison of categorical variables was performed with chi-square or Fisher exact test. Comparison of continuous variables was performed with analysis of variance (ANOVA) or Student *t* test.

EGA indicates estimated gestational age; VA, veno-arterial; VV, veno-venous.

overall mortality rate of this cohort was 49.4%. 360 patients were classified into the After ECMO group, 367 into the On ECMO group, and the remaining 696 into the Mixed group (Fig. 1A). The proportion of patients who actually underwent on-ECMO repair in the After ECMO, Mixed, and On ECMO group was 7.5%, 51.2%, and 88.0%, respectively.

Multivariable Cox Regression

Infants in the On ECMO group were more likely to be inborn and treated at a high-volume center (Table 1). The cumulative incidence of death in the After ECMO, Mixed, and On ECMO groups was 50.4%, 52.7%, and 42.0%, respectively (Table 3). On multivariable Cox regression, compared to the After ECMO group, mortality rate was similar in the Mixed group, hazard ratio (HR) 0.92 (0.74, 1.14), but significantly reduced in the On ECMO group, HR 0.63 (0.48, 0.82) (Table 3).

PS Matching

After PS matching of the On ECMO and After ECMO groups, 136 patients remained in each group and all demographic and baseline characteristics were balanced (Table 1). Kaplan–Meier analysis showed a strong survival advantage with the approach of on-ECMO repair (Fig. 2). Patients in the On ECMO group demonstrated a significantly higher median survival time, 166 versus 53 days ($P < 0.001$). On Cox regression, the On ECMO group again demonstrated a strong survival advantage, with an almost 50% reduction in mortality rate, HR 0.54 (0.38, 0.77) (Table 3). On cluster analysis with a mixed effects logistic regression model, surgical repair On ECMO also lowered the odds of death compared to After ECMO repair, OR 0.85 (0.75, 0.96) (Table 3).

Among the secondary outcomes, patients in the On ECMO group demonstrated more than 5-fold reduction in the incidence of non-repair, 5.9% versus 33.8%, compared to the After ECMO group (Table 3). The On ECMO group also exhibited longer ECMO duration, 11 versus 9 days, and higher proportion of oxygen dependence at discharge, 61.4% versus 37.7% (Table 3). There was no difference in hospitalization duration or proportion of oral feeding at discharge.

Sensitivity Analysis

With non-repairs excluded, the mortality rate predictably improved in all 3 study groups. Notably, there was a reverse in the relationship between on-ECMO repair and mortality. The After ECMO group now demonstrated the lowest cumulative incidence of death, 27.6% compared to 42.8% and 37.6% in the Mixed and On ECMO groups, respectively (Table 3). Compared to the After ECMO group, the Mixed and On-ECMO groups demonstrated a 73% and 47% increase in mortality rate, respectively (Table 3).

Aim 2—Early Versus Late Repair on ECMO

In addition to excluding patients repaired before ECMO or on a second run of ECMO, this aim further excluded 453 patients who

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TABLE 2. Aim 2—Early Versus Late Repair on ECMO.

	Before PS Matching				After PS Matching		
	Early (N = 230)	Mixed (N = 442)	Late (N = 240)	*P*	Early (N = 77)	Late (N = 77)	*P*
Male	128 (55.9%)	244 (55.3%)	146 (60.8%)	0.36	45 (58.4%)	44 (57.1%)	1.00
Race/ethnicity				< 0.001			0.79
White	156 (72.9%)	266 (71.1%)	132 (57.4%)		48 (62.3%)	52 (67.5%)	
Black	20 (9.4%)	37 (9.9%)	56 (24.4%)		11 (14.3%)	10 (13.0%)	
Other	38 (17.8%)	71 (19.0%)	42 (18.3%)		18 (23.4%)	15 (19.5%)	
Inborn	68 (29.6%)	119 (27.0%)	153 (63.8%)	< 0.001	34 (44.2%)	37 (48.0%)	0.75
High-volume center	173 (75.2%)	366 (82.8%)	122 (50.8%)	< 0.001	45 (58.4%)	41 (53.2%)	0.63
Birth weight (kg)	2.97 ± 0.51	3.02 ± 0.52	3.04 ± 0.50	0.36	3.02 ± 0.56	3.03 ± 0.50	
EGA (wk)	37.6 ± 2.0	37.8 ± 1.6	38.1 ± 1.6	0.003	38.0 ± 1.8	37.9 ± 1.6	
Prenatal diagnosis	192 (83.5%)	366 (82.8%)	182 (75.8%)	0.05	52 (67.5%)	58 (75.3%)	0.37
APGAR							
1 min	3.7 ± 2.1	3.9 ± 2.3	3.6 ± 2.2	0.33	3.9 ± 2.2	3.3 ± 2.2	
5 min	5.4 ± 2.1	5.9 ± 2.2	5.6 ± 2.2	0.04	5.7 ± 2.0	5.5 ± 2.2	
Cardiac anomaly			0.82			0.92	
None	170 (73.9%)	331 (74.9%)	185 (77.1%)		59 (76.6%)	57 (74.0%)	
Minor	41 (17.8%)	71 (16.1%)	33 (13.8%)		13 (16.9%)	14 (18.2%)	
Major	19 (8.3%)	40 (9.0%)	22 (9.2%)		5 (6.5%)	6 (7.8%)	
Chromosomal anomaly	14 (6.1%)	29 (6.6%)	11 (4.6%)	0.57	5 (6.5%)	4 (5.2%)	1.00
Left side hernia	173 (76.6%)	350 (80.1%)	188 (79.0%)	0.57	62 (80.5%)	60 (77.9%)	0.84
PHTN degree				0.002			0.96
None	20 (9.4%)	16 (4.1%)	17 (8.1%)		6 (7.8%)	5 (6.5%)	
Mild	16 (7.6%)	18 (4.6%)	23 (10.9%)		9 (11.7%)	11 (14.3%)	
Moderate	88 (41.5%)	144 (37.1%)	77 (36.5%)		29 (37.7%)	28 (36.4%)	
Severe	88 (41.5%)	210 (54.1%)	94 (44.6%)		33 (42.9%)	33 (42.9%)	
ECMO mode				0.18			0.79
VV	17 (7.7%)	41 (9.5%)	29 (12.8%)		7 (9.1%)	9 (11.7%)	
VA	204 (92.3%)	389 (90.5%)	197 (87.2%)		70 (90.9%)	68 (88.3%)	

Demographic and baseline characteristics of the study groups before and after PS matching. Categorical variables are expressed as number and percentage. Continuous variables are expressed as mean ± SD. Comparison of categorical variables was performed with chi-square or Fisher exact test. Comparison of continuous variables was performed with ANOVA or Student *t* test.

EGA indicates estimated gestational age; PHTN, pulmonary hypertension; VA, veno-arterial; VV, veno-venous.

underwent repair after ECMO decannulation and 58 non-repairs who were deemed inappropriate for the analysis. These were non-repairs treated at centers that never repair on ECMO. Nine hundred twelve patients were included in the primary analysis of this aim: 230 in the Early group, 240 in the Late group, and 442 in the Mixed group (Fig. 1B). The overall mortality rate of this cohort was 63.0%. The median time to repair in the Early, Mixed, and Late groups was 2.0 days (interquartile range (IQR) 1.0–4.0 d), 6.0 days (IQR 2.0–9.0 d), and 12.0 days (IQR 7.0–15.5 d), respectively.

Multivariable Cox Regression

Patients in the Late group were more likely to be inborn but less likely to be treated at a high-volume center (Table 2). The Late group incurred the highest incidence of death among all 3 study groups (Table 4). On multivariable Cox regression, mortality rate was lower in the Mixed and Early groups compared to the Late group, HR 0.76 (0.59, 0.98) and 0.67 (0.50, 0.89), respectively (Table 4).

PS Matching

PS matching left 77 patients in each group and resulted in balance of demographic and baseline characteristics (Table 2). The median survival time was significantly longer in the Early group, 135 versus 26 days ($P < 0.001$) (Figure 3). There was a strong increase in survival advantage with the Early repair approach, with a 50% reduction in mortality rate, HR 0.51 (0.33, 0.77) (Table 4). On cluster analysis, the Early group also demonstrated reduced odds of death, OR 0.70 (0.60, 0.80) (Table 4).

Among the secondary outcomes, the approach of Early repair on ECMO was associated with a near 5-fold reduction in the incidence of non-repair, 9.1% versus 44.2% (Table 4). The Late group demonstrated longer median ECMO duration, 15 versus 13 days, and hospitalization duration, 95 versus 54 days, compared to the Early group (Table 4). Oxygen dependence and oral feeding showed no difference between the 2 groups.

Sensitivity Analysis

Similar to aim 1, exclusion of non-repairs improved mortality in all 3 study groups. The cumulative incidence of death in the Late, Mixed, and Early groups was 64.3%, 46.8%, and 43.6%, respectively (Table 4). However, the Cox regression model indicated no difference in mortality rate among the 3 study groups (Table 4).

DISCUSSION

This study demonstrated that the approach of surgical repair of CDH on ECMO improves survival compared to off-ECMO repair. Furthermore, early repair on ECMO also decreases mortality compared to delayed repair on ECMO. Both aims of this study showed that the association of survival advantage and the early approach to repair was the result of a significant reduction in the rate of non-repair, after controlling for center and patient-specific baseline characteristics.

This study employed a unique approach to study group classification. As opposed to stratifying study subjects based on their actual timing of repair, patients were classified based on 2 center-level characteristics as an effort to include non-repairs. Infants that go unrepaired, which results in 100% mortality, make up a significant portion of CDH patients and heavily influence the mortality rate. Omitting these patients would result in a significant

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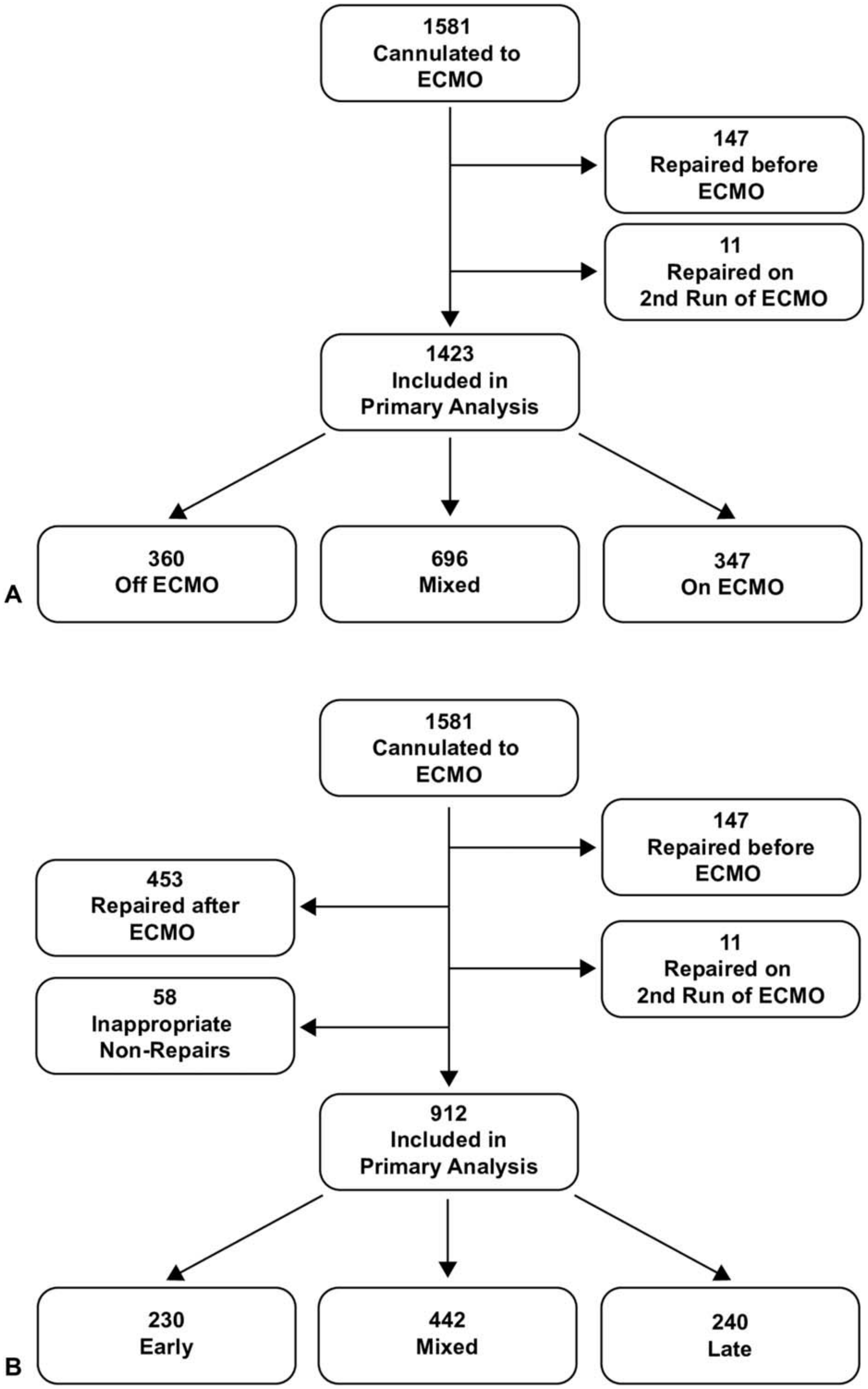

FIGURE 1. Study cohort. Flow chart of patients included in the primary analysis of aim 1 (A) and aim 2 (B).

selection bias that would be very challenging to account for with statistical methods. By allowing for inclusion of non-repairs, we provided a more valid comparison between the approach of early versus delayed repair after ECMO cannulation.

After non-repairs were excluded in aim 1, the relationship between surgical timing and mortality reversed. In addition, the HR of 1.47 for the On ECMO cohort, compared to the After ECMO cohort following non-repair exclusion, is strikingly similar to that reported by Bryner et al[10] despite different methods for study group classification. This sensitivity analysis demonstrated the profound influence of non-repairs on the relationship between surgical timing and mortality. Surgical repair of CDH after ECMO decannulation

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TABLE 3. Aim 1—On Versus After ECMO Repair.

	Multivariable Cox Regression				PS Matching			Sensitivity Analysis			
	After ECMO	Mixed	On ECMO	P	After ECMO	On ECMO	P	After ECMO	Mixed	On ECMO	P
Primary outcome											
Death	181/359 (50.4%)	366/694 (52.7%)	153/364 (42.0%)		75/136 (55.2%)	53/136 (39.0%)		68/246 (27.6%)	244/570 (42.8%)	127/338 (37.6%)	
HR*	Ref	0.92 (0.74, 1.14)	0.63 (0.48, 0.82)	0.001	Ref	0.54 (0.38, 0.77)	< 0.001	Ref	1.73 (1.30, 2.30)	1.47 (1.06, 2.02)	0.001
OR†					Ref	0.52 (0.32, 0.85)	0.01				
Secondary outcomes											
Non-repair					46/136 (33.8%)	8/136 (5.9%)	< 0.001				
ECMO duration (d)					9 (6, 15)	11 (8, 17)	0.001				
Hospitalization duration (d)‡					71 (51, 103)	68 (45, 98)	0.62				
O2 dependence at discharge‡					23/61 (37.7%)	51/83 (61.4%)	0.005				
Oral feeding at discharge‡					20/61 (32.8%)	27/80 (33.8%)	0.90				

Analyses of outcomes with standard multivariable Cox proportional hazard regression, PS matching, and sensitivity analysis. In sensitivity analysis, multivariable Cox regression was repeated after exclusion of non-repairs. Dichotomous outcomes are expressed as number and percentage. Continuous outcomes are expressed as median and interquartile range (IQR).

*Hazard ratio (HR) for mortality was calculated from a Cox proportional hazard regression model. HRs are displayed with 95% confidence interval (CI).

†Odds ratio (OR) for death was calculated from a mixed effects logistic regression model, taking into account clustering by treatment centers and allowing the intercept of each center to have a random effect. OR is displayed with 95% CI.

‡Hospitalization duration, oxygen dependence, and oral feeding were only calculated among survivors.

per se would probably result in more favorable outcomes, possibly due to more stable hemodynamic status, absence of systemic anticoagulation,[5,17,18] and elimination of patients who would not be rescued by either intervention. However, the approach of waiting and repairing after ECMO decannulation would leave more patients unrepaired, and thus result in more non-survivors, which largely outweighs the potential physiologic benefits of off-ECMO repair. Further, predicting an individual patient's ability to be liberated from ECMO at the beginning of support is fraught with uncertainty.

In aim 2, non-repairs again greatly influenced the relationship between surgical timing and mortality, as their exclusion neutralized the survival advantage associated with early repair. Although not captured in our analyses, other factors may also contribute to improved survival with early repair on ECMO, such as reduced edema and greater ease of surgical correction. Results of aim 2 therefore further strengthened the conclusion of aim 1. While the timing of surgery itself may have some influence on mortality, the ultimate survival advantage belongs to the surgical approach that minimizes the rate of non-repair. In both aims, this approach is early repair after ECMO cannulation.

Of note, most patients ($>$ 70%) in the On ECMO group in aim 1 and Early group in aim 2 were treated at high-volume centers. This suggests that high-volume centers tend to be more ''aggressive'' with regard to surgical repair timing after ECMO cannulation. This is consistent with a prior study from the CDHSG registry where ''aggressive'' centers were shown to have significantly lower risk-adjusted mortality.[19] Early repair may be the key component of ''aggressive'' surgical management and accountable for the lower rate of non-repair and death. In order to account for this potential confounder and to ensure nonbiased assessment of surgical timing approach on mortality, high-volume status therefore was included in the multivariable Cox regression model as well as PS matching. Additionally, in order to improve the inherent lack of equipoise in this retrospective study, we also performed a mixed effects logistic regression model to account for the clustering effect of treatment centers. In both aims, the fact that early repair of CDH was associated with survival advantage across different statistical models confirmed the validity of our findings.

This study also produced some noteworthy secondary outcomes. One of them is the higher proportion of oxygen dependence at discharge with on-ECMO repair. This could be explained from a higher proportion of high-risk infants being salvaged with the early repair approach. In other single-institutional studies, pulmonary support at 30 days and discharge has been associated with neurodevelopmental delay and increased long-term oxygen requirement.[20,21] As the CDHSG registry only collects data until discharge from index hospitalization, long-term outcome is lacking in this database and further multi-institutional studies are needed to confirm this relationship. Another interesting secondary outcome was the relationship between surgical timing and duration of hospitalization. Early repair on ECMO significantly reduced the median length of hospitalization compared to late repair (54 vs 95 d). Given that CDH is among the costliest pediatric surgical conditions,[22,23] this degree of hospital day reduction could provide additional impetus for the approach of early repair after ECMO cannulation.

Results of this study need to be taken in the context of its design. The retrospective and observational nature of the study increased the probability of bias. Despite the use of multivariable regression models and PS matching, it is possible that certain confounders were left unmeasured or unadjusted for. One important confounder that was not included in the regression or PS model was CDHSG defect size, a marker of pulmonary hypoplasia. Since defect size can only be assessed at the time of surgical repair and this study was designed to include all non-repairs, the high degree of missing data precluded its

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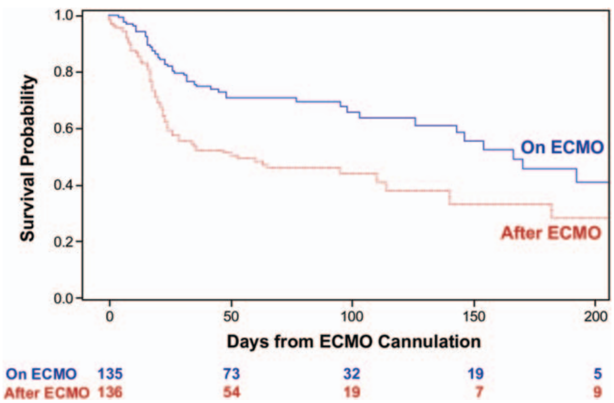

FIGURE 2. Aim 1—On versus After ECMO Repair. Kaplan–Meier curves for patients in the On ECMO and After ECMO groups after propensity score (PS) matching.

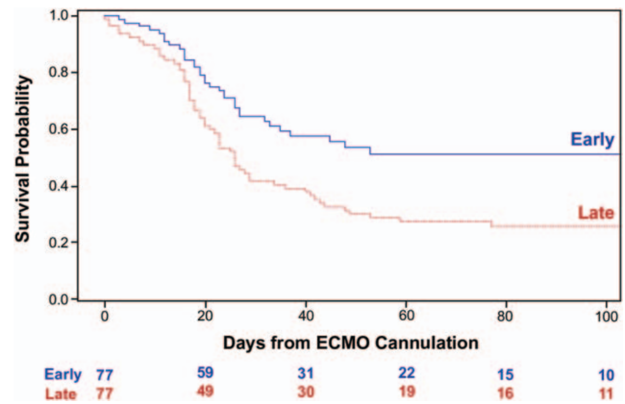

FIGURE 3. Aim 2—Early versus Late Repair on ECMO. Kaplan–Meier curves for patients in the Early and Late groups after PS matching.

TABLE 4. Aim 2—Early Versus Late Repair on ECMO.

	Multivariable Cox Regression				PS Matching			Sensitivity Analysis			
	Late	Mixed	Early	*P*	Late	Early	*P*	Late	Mixed	Early	*P*
Primary outcome											
Death	188/239 (78.7%)	247/439 (56.3%)	115/229 (50.2%)		62/77 (80.5%)	34/77 (44.2%)		92/143 (64.3%)	167/357 (46.8%)	88/202 (43.6%)	
HR*	*Ref*	0.76 (0.59, 0.98)	0.67 (0.50, 0.89)	0.02	*Ref*	0.51 (0.33, 0.77)	0.002	*Ref*	0.88 (0.64, 1.20)	0.82 (0.59, 1.15)	0.52
OR†					*Ref*	0.19 (0.09, 0.39)	< 0.001				
Secondary outcomes											
Non-repair					34/77 (44.2%)	7/77 (9.1%)	< 0.001				
ECMO duration (d)					15 (12, 23)	13 (8, 19)	0.02				
Hospitalization duration (d)‡					95 (82, 145)	54 (33, 91)	0.004				
O2 dependence at discharge‡					7/15 (46.7%)	28/43 (65.1%)	0.21				
Oral feeding at discharge‡					1/15 (6.7%)	10/40 (25.0%)	0.25				

Analyses of outcomes with standard multivariable Cox proportional hazard regression, PS matching, and sensitivity analysis. In sensitivity analysis, multivariable Cox regression was repeated after exclusion of non-repairs. Dichotomous outcomes are expressed as number and percentage. Continuous outcomes are expressed as median and IQR.

*HR for mortality was calculated from a Cox proportional hazard regression model. HRs are displayed with 95% confidence interval (CI).

†OR for death was calculated from a mixed effects logistic regression model, taking into account clustering by treatment centers and allowing the intercept of each center to have a random effect. OR is displayed with 95% CI.

‡Hospitalization duration, oxygen dependence, and oral feeding were only calculated among survivors.

inclusion in the statistical models. However, defect size was implicit in other included covariates such as prenatal diagnosis, Apgar score, and the degree of pulmonary hypertension. With the use of a voluntary multi-institutional registry, there are some outcomes that failed to be captured. This can include bleeding and other complications on ECMO, which would provide additional insight into the impact of surgical timing approach on ultimate patient outcome.

Despite these potential shortcomings, our results demonstrate that the approach of early repair on ECMO results in significantly lower rate of non-repair and, consequently, mortality. Although factors such as the patient's clinical status and ECMO experience of individual treatment centers should be taken into account, centers should attempt to repair CDH early after ECMO cannulation as an effort to improve survival.

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REFERENCES

1. Nobuhara KK, Wilson JM. Pathophysiology of congenital diaphragmatic hernia. *Semin Pediatr Surg*. 1996;5:234–242.
2. Colvin J, Bower C, Dickinson JE, et al. Outcomes of congenital diaphragmatic hernia: a population-based study in Western Australia. *Pediatrics*. 2005;116:e356–e363.
3. Bent DP, Nelson J, Kent DM, et al. Population-based validation of a clinical prediction model for congenital diaphragmatic hernias. *J Pediatr*. 2018;201:160–165.e1.

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4. Kays DW. ECMO in CDH: is there a role? *Semin Pediatr Surg*. 2017;26:166–170.
5. Thiagarajan RR, Barbaro RP, Rycus PT, et al. Extracorporeal Life Support Organization Registry International Report 2016. *ASAIO J*. 2017;63:60–67.
6. Turek JW, Nellis JR, Sherwood BG, et al. Shifting risks and conflicting outcomes-ECMO for neonates with congenital diaphragmatic hernia in the modern era. *J Pediatr.* 2017;190:163–168.e4.
7. Lally KP, Paranka MS, Roden J, et al. Congenital diaphragmatic hernia. Stabilization and repair on ECMO. *Ann Surg*. 1992;216:569–573.
8. Fallon SC, Cass DL, Olutoye OO, et al. Repair of congenital diaphragmatic hernias on Extracorporeal Membrane Oxygenation (ECMO): does early repair improve patient survival? *J Pediatr Surg*. 2013;48:1172–1176.
9. Robertson JO, Criss CN, Hsieh LB, et al. Comparison of early versus delayed strategies for repair of congenital diaphragmatic hernia on extracorporeal membrane oxygenation. *J Pediatr Surg*. 2018;53:629–634.
10. Congenital Diaphragmatic Hernia Study Group BS, Bryner BS, West BT, et al. Congenital diaphragmatic hernia requiring extracorporeal membrane oxygenation: does timing of repair matter? *J Pediatr Surg*. 2009;44:1165–1171.
11. Lally KP, Lasky RE, Lally PA, et al. Standardized reporting for congenital diaphragmatic hernia—an international consensus. *J Pediatr Surg*. 2013;48:2408–2415.
12. Guner YS, Harting MT, Fairbairn K, et al. Outcomes of infants with congenital diaphragmatic hernia treated with venovenous versus venoarterial extracorporeal membrane oxygenation: a propensity score approach. *J Pediatr Surg*. 2018;53:2092–2099.
13. Zani-Ruttenstock E, Zani A, Eaton S, et al. First population-based report of infants with congenital diaphragmatic hernia: 30-day outcomes from the American College of Surgeons National Quality Improvement Program. *Eur J Pediatr Surg*. 2019;29. 62–67.
14. Burgos CM, Frenckner B, Luco M, et al. Prenatally versus postnatally diagnosed congenital diaphragmatic hernia—side, stage, and outcome. *J Pediatr Surg*. 2019;54. 651–655.
15. Puligandla PS, Grabowski J, Austin M, et al. Management of congenital diaphragmatic hernia: a systematic review from the APSA outcomes and evidence based practice committee. *J Pediatr Surg*. 2015;50:1958–1970.
16. Brindle ME, Cook EF, Tibboel D, et al. A clinical prediction rule for the severity of congenital diaphragmatic hernias in newborns. *Pediatrics*. 2014;134:e413–e419.
17. Vazquez WD, Cheu HW. Hemorrhagic complications and repair of congenital diaphragmatic hernias: does timing of the repair make a difference? Data from the Extracorporeal Life Support Organization. *J Pediatr Surg*. 1994;29:1002–1005.
18. Golden J, Jones N, Zagory J, et al. Outcomes of congenital diaphragmatic hernia repair on extracorporeal life support. *Pediatr Surg Int*. 2017;33:125–131.
19. Harting MT, Hollinger L, Tsao K, et al. Aggressive surgical management of congenital diaphragmatic hernia. *Ann Surg*. 2017;267:1.
20. Cortes RA, Keller RL, Townsend T, et al. Survival of severe congenital diaphragmatic hernia has morbid consequences. *J Pediatr Surg*. 2005;40:36–45.
21. Cauley RP, Potanos K, Fullington N, et al. Pulmonary support on day of life 30 is a strong predictor of increased 1 and 5-year morbidity in survivors of congenital diaphragmatic hernia. *J Pediatr Surg*. 2015;50:849–855.
22. Cameron DB, Graham DA, Milliren CE, et al. Quantifying the burden of interhospital cost variation in pediatric surgery. *JAMA Pediatr.* 2017;171:e163926.
23. Faraoni D, Nasr VG, DiNardo JA, et al. Hospital costs for neonates and children supported with extracorporeal membrane oxygenation. *J Pediatr.* 2016;169:69–75.e1.

Appendix. Participating centers in the Congenital Diaphragmatic Hernia Study Group Registry

Center	City	State/ Province	Country
Alberta Children's Hospital	Calgary	AB	Canada
Arkansas Children's Hospital	Little Rock	AR	USA
Astrid Lindgren Children's Hospital	Stockholm		Sweden
Azienda Ospedaliera Papa Giovanni XXIII	Bergamo		Italy
BC Children's & Women's Health Centre	Vancouver	BC	Canada
Cairo University Pediatric Hospital (Aboul Reesh)	Cairo		Egypt
Carolinas Medical Center, Levine Children's Hospital	Charlotte	NC	USA
Children's Hospital & Research Center Oakland	Oakland	CA	USA
Childrens Hospital at Skanes University Hospital	Lund		Sweden
Children's Hospital Boston	Boston	MA	USA
Children's Hospital of Akron	Akron	OH	USA
Children's Hospital of Georgia—AU Health	Augusta	GA	USA
Children's Hospital of Illinois at OSF St. Francis Med Center	Peoria	IL	USA
Children's Hospital of Los Angeles	Los Angeles	CA	USA
Childrens' Hospital of Orange County	Orange	CA	USA
Children's Hospital of San Antonio	San Antonio	TX	USA
Children's Hospital of Wisconsin	Milwaukee	WI	USA
Children's Hospital Omaha	Omaha	NE	USA
Childrens Hospital, University Bonn	Bonn		Germany
Children's Hospitals and Clinics (Minneapolis)	Minneapolis	MN	USA
Children's Memorial Hermann Hospital	Houston	TX	USA
Children's of Alabama	Birmingham	AL	USA
Cincinnati Children's Hospital Medical Center	Cincinnati	OH	USA
Cleveland Clinic Foundation-Children's Hospital	Cleveland	OH	USA
Connecticut Children's Medical Center	Hartford	CT	USA
Dell Children's Medical Center of Central Texas	Austin	TX	USA
Duke University Medical Center	Durham	NC	USA
Emory University	Atlanta	GA	USA
Golisano Children's Hospital at Strong	Rochester	NY	USA
Hospital Clinico Universidad Católica de Chile	Santiago	RM	Chile
IRCCS Fondazione Ca' Granda Ospedale Maggiore Policlinico	Milano		Italy
James Whitcomb Riley Children's Hospital	Indianapolis	IN	USA
Johns Hopkins All Children's Hospital	St Petersburg	FL	USA
Johns Hopkins Hospital	Baltimore	MD	USA
Juan P. Garrahan Children Hospital	Buenos Aires		Argentina
Le Bonheur Children's Medical Center	Memphis	TN	USA
Legacy Emanuel Children's Hospital	Portland	OR	USA

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Appendix (*Continued*)

Center	City	State/ Province	Country
Loma Linda University Children's Hospital	Loma Linda	CA	USA
Lucile Salter Packard Children's Hospital	Palo Alto	CA	USA
Mattel Children's Hospital at UCLA	Los Angeles	CA	USA
Miami Valley Hospital	Dayton	OH	USA
National Center for Child Health and Development	Setagaya-ku	Tokyo	Japan
NICU Health Sciences Centre	Winnipeg	MB	Canada
Norton Children's Hospital	Louisville	KY	USA
Osaka University Graduate School of Medicine	Suita	Osaka	Japan
Ospedale Pediatrico Bambino Gesù	Rome		Italy
Palmetto Health Richland	Columbia	SC	USA
Phoenix Children's Hospital	Phoenix	AZ	USA
Polish Mother's Memorial Hospital Research Institute	Lodz		Poland
Primary Children's Hospital	Salt Lake City	UT	USA
Radboud University Nijmegen Medical Centre	Nijmegen		The Netherlands
Rady Children's Hospital	San Diego	CA	USA
Research Center for Obstetrics, Gynecology and Perinatology	Moscow		Russia
Research Institute at Nationwide Children's Hospital	Columbus	OH	USA
Royal Children's Hospital	Parkville	Victoria	Australia
Royal Hospital for Sick Children	Glasgow		Scotland
Shands Children's Hospital/ University of Florida	Gainesville	FL	USA
Sophia Children's Hospital	Rotterdam		The Netherlands
St. Francis Children's Hospital	Tulsa	OK	USA
St. Joseph's Hospital and Medical Center	Phoenix	AZ	USA

Appendix (*Continued*)

Center	City	State/ Province	Country
St. Louis Children's Hospital	St. Louis	MO	USA
St. Louis Univ School of Medicine at SSM Health Cardinal Glennon Children's Hospital	St. Louis	MO	USA
Stollery Children's Hospital	Edmonton	AB	Canada
Sydney Children's Hospital	Randwick	NSW	Australia
Texas Children's Hospital	Houston	TX	USA
The Children's Hospital at OU Medical Center	Oklahoma City	OK	USA
The Children's Hospital of Pittsburgh of UPMC	Pittsburgh	PA	USA
The Hospital for Sick Children	Toronto	Ontario	Canada
The Queen Silvia Children's Hospital SU/Östra	Gothenburg		Sweden
Tufts Medical Center	Boston	MA	USA
UNC School of Medicine	Chapel Hill	NC	USA
University Childrens Hospital	Uppsala		Sweden
University Malaya Medical Centre	Kuala Lumpur		Malaysia
University of Michigan, C.S. Mott Children's Hospital	Ann Arbor	MI	USA
University of Nebraska Medical Center	Omaha	NE	USA
University of Padua	Padua		Italy
University of Texas Medical Branch at Galveston	Galveston	TX	USA
University of Virginia Medical School	Charlottesville	VA	USA
Vanderbilt Children's Hospital	Nashville	TN	USA
Vladivostok State Medical University	Vladivostok		Russia
Winnie Palmer Hospital for Women & Babies	Orlando	FL	USA
Yale New Haven Children's Hospital	New Haven	CT	USA

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